

Shanmukesh Bonala

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[Portfolio](#) — [GitHub](#) — [LinkedIn](#)

PROFESSIONAL SUMMARY

High-achieving CS honors student (**Top 10%, 9.35 CGPA**) and AI/ML Researcher specializing in Trustworthy AI, sustainable Deep Learning, and Computer Vision. Track record includes building multi-modal forensic classifiers, open-source multi-agent networks, and deploying optimized edge architectures under rigid latency constraints. Bridges the gap between rigorous research and production engineering by combining theoretical ML expertise with robust backend skills.

EDUCATION

VIT-AP University

B.Tech, Computer Science and Engineering (AI-ML) — **CGPA: 9.35 (Top 10% in Dept.)**

2023 – 2027

Amaravati, India

Bheeram Sreedhar Reddy Jr. College

2021 – 2023

Intermediate (MPC) — **97.3%**

Kadapa, India

TECHNICAL SKILLS

- **AI Frameworks & Tools:** PyTorch, OpenCV, Hugging Face, IBM Watsonx.
- **CV & Deep Learning:** Image Forensics (DWT/FFT/ELA), YOLOv8, CNNs (EfficientNet, ConvNeXt), Multi-Modal Systems.
- **Generative AI & Agents:** RAG Pipelines, Multi-Agent Orchestration, Prompt Engineering.
- **Web Technologies:** ReactJS, Next.js, HTML5, CSS3, JavaScript (ES6+), TypeScript.
- **Backend & Data:** Python (FastAPI), SQL, Supabase (PostgreSQL), Real-time Pub/Sub.
- **Tools & Cloud:** Docker, Git, WSL, Google Cloud Platform (GCP Cloud Run), Firebase, Vercel.
- **Core Concepts:** Data Structures & Algorithms, RESTful APIs, Clean Code, Agile Methodology.

EXPERIENCE

VNIT-Nagpur

June 2026

Research Intern (Under Dr. Anil Mokhade, Associate Professor)

- Trained deep neural variants to optimize predictive throughput, yielding a **14% increase in FPS** while executing comparative ablation studies to reduce overall parameter overhead.
- Profiled computational bottlenecks across architectural paradigms, deploying custom structural pruning configurations that achieved a **30% parameter reduction** and cut inference latency by **22ms** with strict stability.

PROJECTS

E2AM: Energy-Aware Multi-Model Workspace (Green AI)

Feb 2026 – July 2026

- **Solely designed and executed** benchmarking of 4 vision models (EfficientNetV2-S, ResNet50, ConvNeXtV2-Tiny, MobileViT-V2) across CIFAR-10/100 and Tiny-ImageNet to systematically evaluate performance against hardware compute costs.
- Orchestrated 684+ GPU hours across 225 unique configurations and 11,250 epochs, implementing strict carbon-tracking mechanisms and efficiency-first reproducibility metrics.
- Created and published a Python library for reproducible benchmarking workflows, available via `pip install e2am`.

AutoSource: Autonomous Multi-Agent Procurement Network

June 2026 – Present

- **Architected and independently developed** an open-source autonomous multi-agent ecosystem for SMEs that automates real-time inventory monitoring, vendor sourcing, and algorithmic multi-turn price negotiations.
- Implemented reliable human-in-the-loop validation protocols and modular agent workflows to ensure end-to-end supply chain transparency and measurable cost reductions.

HybridForensicsNetV3: Multi-Modal AI-Generated Image Detection

Nov 2025 – Jan 2026

- **Served as core developer** for a data-centric image forensics classifier yielding a **99.5% F1-score** and **100% Recall** in identifying synthetic, AI-generated image distributions.
- Executed comprehensive ablation studies across multi-modal inputs (RGB, FFT, DWT, ELA) to eliminate noise-heavy feature corruption, proving that streamlined RGB baselines optimized runtime efficiency without compromising safety.

LDW-CNet: Interpretable Medical Image Classification

Aug 2025 – Oct 2025

- **Co-authored and engineered** a lightweight CNN framework incorporating a Learnable Discrete Wavelet Transform (DWT) layer to produce highly interpretable brain tumor classifications from MRI scans.
- Enforced custom constraint loss functions during network training to optimize architectural constraints and structural convergence.

Urban Traffic Management System

Jan 2026

- **Developed as a personal project** a responsive React admin dashboard featuring high-frequency telemetry charting tools to monitor real-time spatial data streams.
- Designed an edge-to-cloud Pub/Sub data tier that eliminates data drift and maintains consistent state synchronization under low-latency limits (<200ms).

PROFESSIONAL ACTIVITIES & ACHIEVEMENTS

- **Professional Affiliations:** Student Member of the IEEE; Active Student Reviewer for selected Elsevier journals.
- **Hackathons:** Finalist at the Google Hackathon (Smart Campus Dashboard) and MSME Hackathon (Smart TrackX).
- **Certifications:** [AWS Certified AI Practitioner](#) — [AWS Certified Cloud Practitioner](#) — [IBM Generative AI Dev \(Watsonx\)](#) — [Apna College Delta 2.0 \(Full-Stack\)](#)